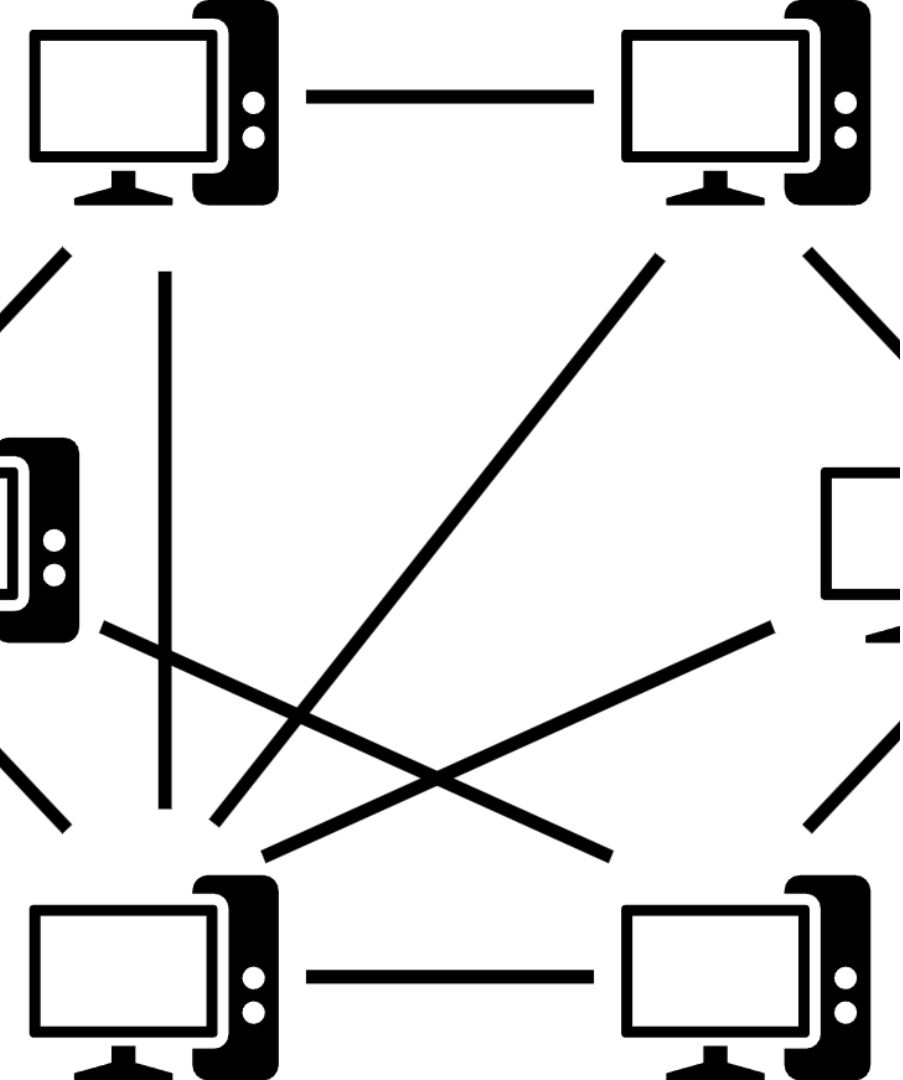




Tanglement.ai: Decentralizing AI Infrastructure

An introduction to Tanglement.ai, a peer-to-peer routing layer for affordable and reliable access to large language models.

Team & Advisors



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The LLM Access Crisis



High Costs

\$0.002-0.03 per token with NO optimization, organizations pay full retail rates, no volume discounts or intelligent routing



Vendor Lock-In

Single provider dependency, no failover when providers go down, forced acceptance of price increases



Unpredictable Reliability

Frequent service outages, production apps disrupted by downtime, no SLA guarantees for most users



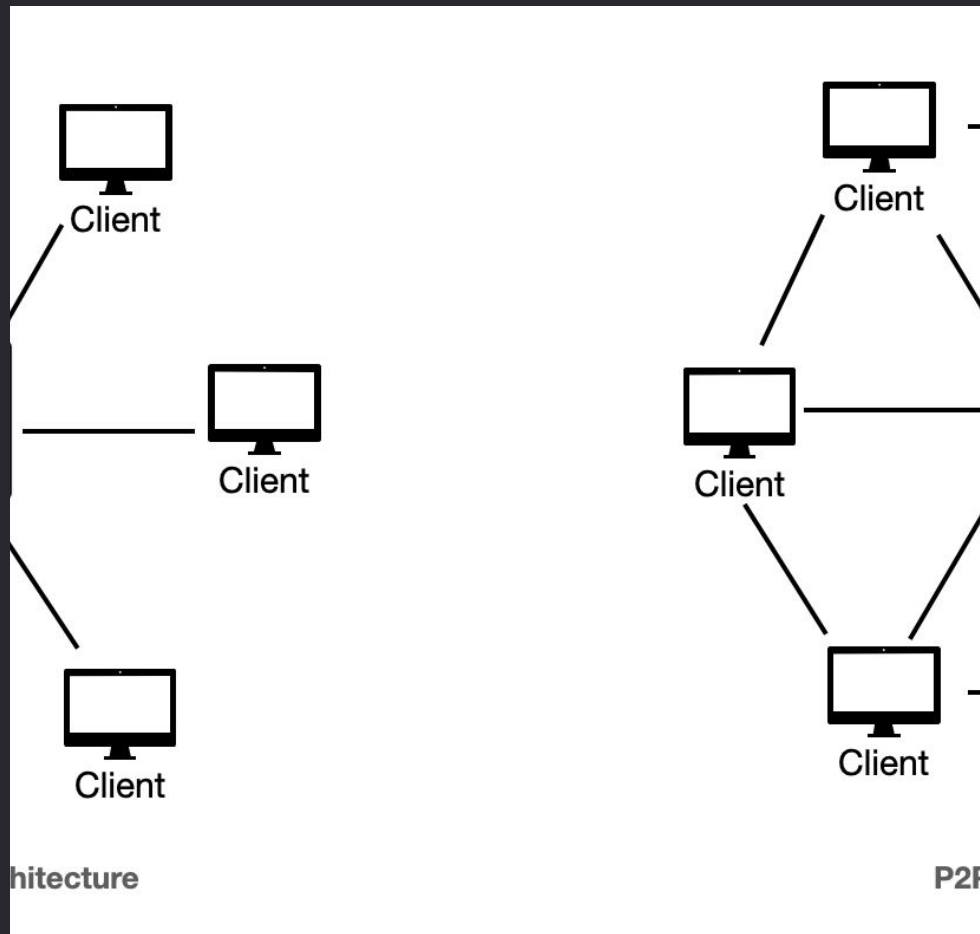
Zero Cost Transparency

Hidden pricing changes and opaque billing, unexpected charges, surge pricing without notice

The current centralized model of LLM access is plagued by high costs, vendor lock-in, unreliable service, and a lack of transparency, creating a crisis for organizations that rely on these critical AI tools.

Tanglement.ai's Solution

Tanglement.ai's peer-to-peer routing layer eliminates the need for centralized infrastructure, solving the LLM access crisis. By leveraging proven distributed systems technologies, Tanglement.ai enables intelligent client-side routing to LLM providers without a middleman. This architecture achieves zero bottlenecks, zero single points of failure, and significantly reduced operational costs compared to traditional API gateways.



Token Economics

- What Tanglement.ai Is

✓ Credits for LLM access through the network ✓ Rewards for contributing compute/bandwidth/storage ✓ Fixed supply: 1 billion tokens

- What Tanglement.ai Is NOT

✗ NOT a cryptocurrency (no trading/speculation) ✗ NOT a governance token ✗ NOT listed on exchanges

- Token Acquisition

1. Purchase with fiat (USD/EUR via payment processor) 2. Earn through contribution (CPU, bandwidth, storage, uptime) 3. Ecosystem grants (developers building on platform) 4. Promotional rewards (early adopter incentives)

- Token Distribution (1B Total)

40% - Contribution mining rewards 30% - Ecosystem development 20% - Team (4-year vesting, 1-year cliff) 10% - Initial liquidity

- Economic Flow

Premium Users (1.2× rate) ↓ Transaction Fee (0.5-2%) ↓ Subsidy Pool ↓ Economy Users (0.6-0.8× rate)

- Contributors Earn Tokens By Providing

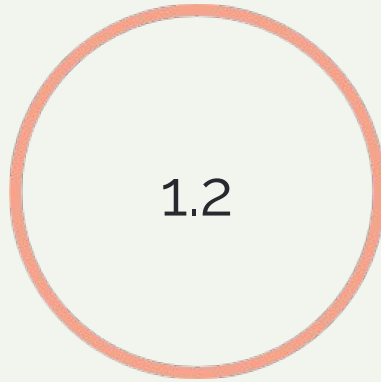
• CPU time (network processing) • Bandwidth (request relaying) • Storage (routing table hosting) • Uptime (online & reachable)

Three-Tier Value Proposition

Premium Reliability, Premium Performance, and Economy Pricing



Premium Reliability



Premium Performance



Economy Pricing

Competitive Advantage

Category	Tanglement.ai	Traditional Gateways	Direct Provider
Architecture	✓ Fully Distributed P2P	⚠ Centralized Servers	✗ Provider-Owned
Operational Costs	✓ \$357/month	⚠ \$100k+/month	N/A
Multi-Provider Routing	✓ Client-Sidentelligence	⚠ Limited	✗ None
Cost Savings	✓ 20-40% (Economy)	⚠ 5-10%	✗ 0%
Privacy Model	✓ Zero-Knowledge	⚠ Logs Everything	⚠ Provider Logs
Vendor Lock-in	✓ None	⚠ Medium	✗ High
Single Point of Failure	✓ None (P2P)	✗ High	⚠ Provider Outages
Censorship Resistance	✓ High	✗ Low	✗ Provider Control

Technical Architecture

DHT (Chord Protocol)

- $O(\log N)$ peer discovery • Scales to 100k+ nodes • Consistent hashing for load balance

Gossip Protocol

- Epidemic state synchronization • Fanout=6, <5min convergence • Routing table propagation

WireGuard Mesh

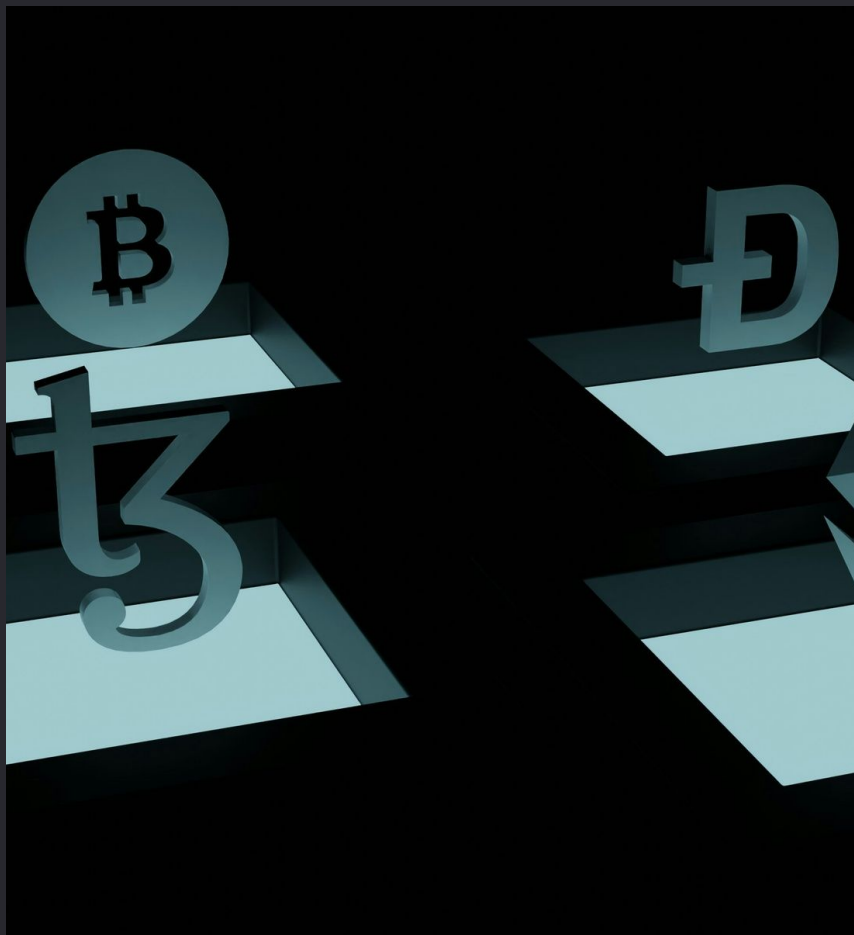
- Encrypted P2P connections • ~5ms per-hop latency • Minimal CPU overhead (~1-2%)

Signal Protocol

- End-to-end encryption • Forward secrecy • Zero-trust architecture

Blockchain/IPFS Storage

- Routing table stored on-chain • Censorship resistant • Global availability



Revenue Model

Tanglement.ai's revenue model is based on two main streams: primary revenue from token sales and secondary revenue from premium services. The primary revenue comes from users purchasing TAI tokens to access the LLM routing network, with a three-tier pricing model that creates market segmentation. The premium tiers, which offer enhanced reliability and performance, help fund the subsidized economy tier. Additionally, Tanglement.ai generates secondary revenue from optional premium services such as PII/PHI detection, prompt injection prevention, content moderation, and compliance reporting.

Go-to-Market Strategy

● Phase 1: Early Adopters
100 active organizations, 1,000 network nodes, proven P2P reliability

● Phase 2: Enterprise Expansion

50 enterprise customers, 10,000 network nodes, \$1M ARR from premium services

● Phase 3: Geographic Expansion
100,000+ nodes globally, 5+ regions covered, \$10M ARR

● Phase 4: Platform Maturation
Self-sustaining network economy, 1,000+ organizations, platform profitability

Success Metrics

Technical KPIs

- Latency: P95 <1s routing overhead - Throughput: 10M RPS sustained at 10k nodes - Availability: 99.9% uptime (Premium), 95%+ (Economy) - Decentralization: 95%+ requests routed without company infrastructure

Economic KPIs

- Cost Savings: 25% average, 40% for Economy tier - Token Velocity: 10+ transactions per token per month - Premium Service Revenue: \$1M ARR by Year 1 - Network Revenue: \$10M ARR by Year 2 - Contribution Rate: 80% of nodes actively contributing

Growth KPIs

- Node Growth: 10k nodes by Month 12 - Request Volume: 100M requests/month by Month 12 - User Acquisition: 1,000 organizations by Month 18 - Geographic Coverage: 5+ regions by Month 24

Current Traction

[Fill in actual numbers if available, otherwise remove this section]

Risk Management



Provider Terms of Service Violation

LLM providers prohibit credential sharing/resale in ToS. Mitigate by having users provide their own API keys, direct client-provider calls, formal provider partnerships, and legal compliance review.



Client Reverse Engineering

Attackers could reverse engineer SDK to extract keys or exploit network. Mitigate with LLVM-based obfuscation, binary packing, integrity checks, white-box cryptography, and frequent client updates.



Token Securities Regulation

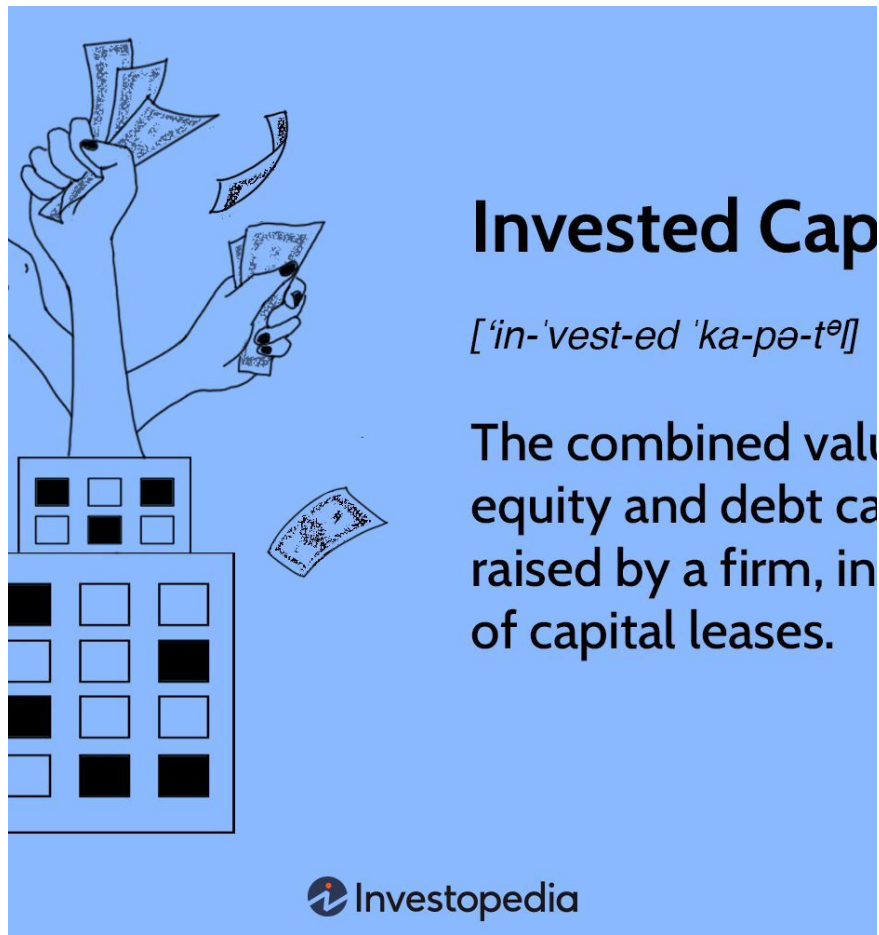
Risk of TAI token being classified as a security, requiring SEC registration. Mitigate with pure utility token design, no secondary trading, proactive regulatory engagement, and legal counsel specialization.



Byzantine Node Attacks

Malicious nodes broadcasting false routing data. Mitigate with peer attestation, proof-of-stake requirements, slashing for misbehavior, client-side validation, and centralized fallback for high Byzantine ratio.

Tanglement.ai proactively addresses major risks through a combination of technical, operational, and legal strategies to ensure the long-term stability and security of the decentralized network.



The Ask

Tanglement.ai is seeking \$10M in Series A funding to scale its decentralized LLM infrastructure. The funding will be used to expand the engineering team, strengthen legal and compliance efforts, and drive go-to-market initiatives. With this investment, Tanglement.ai aims to reach 10,000 active nodes and \$1M in annual recurring revenue within 18 months, paving the way for long-term global expansion and community governance.

Understanding the Challenge in Blockchain AI

The Problems Without Phala's TEE Solution

Centralized
em



**AI Model Training
Needs Program State
Secrecy**



Access
Vulnerability
Resilience

AI Model
Corruption

Tanglement.ai is building the decentralized routing layer that will power the next generation of affordable, censorship-resistant, and privacy-preserving AI applications. By eliminating the need for centralized infrastructure, Tanglement.ai will become the standard for accessing large language models, enabling a more open and equitable AI ecosystem. Through its innovative peer-to-peer architecture and token-based model, Tanglement.ai is poised to disrupt the current LLM access crisis and pave the way for a future where AI is truly accessible to all.